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Semiconductor device structure and methods of forming the same

Abstract

A semiconductor device structure, along with methods of forming such, are described. The semiconductor device structure includes a device, a first conductive structure disposed over the device, and the first conductive structure includes a first sidewall having a first portion and a second portion. The semiconductor device structure further includes a first spacer layer disposed on the first portion, a second conductive structure disposed adjacent the first conductive structure, and the second conductive structure includes a second sidewall having a third portion and a fourth portion. The semiconductor device structure further includes a second spacer layer disposed on the third portion, and an air gap is formed between the first conductive structure and the second conductive structure. The second portion, the first spacer layer, the fourth portion, and the second spacer layer are exposed to the air gap.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. patent application Ser. No. 17/723,427 filed Apr. 18, 2022, which is a divisional

BACKGROUND

(1) As the semiconductor industry introduces new generations of integrated circuits (IC) having higher performance and more functionality, the density of the elements forming the ICs increases, while the dimensions, sizes and spacing between components or elements are reduced. In the past, such reductions were limited only by the ability to define the structures photo-lithographically, device geometries having smaller dimensions created new limiting factors. For example, for any two adjacent conductive features, as the distance between the conductive features decreases, the resulting capacitance (a function of the dielectric constant (k value) of the insulating material divided by the distance between the conductive features) increases. This increased capacitance results in increased capacitive coupling between the conductive features, increased power consumption, and an increase in the resistive-capacitive (RC) time constant.

(2) Therefore, there is a need to solve the above problems.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 is a perspective view of one of the various stages of manufacturing a semiconductor device structure, in accordance with some embodiments.

(3) FIGS. 2A-22A are cross-sectional side views of various stages of manufacturing the semiconductor device structure taken along line A-A of FIG. 1, in accordance with some embodiments.

(4) FIGS. 2B-22B are cross-sectional side views of various stages of manufacturing the semiconductor device structure taken along line B-B of FIG. 1, in accordance with some embodiments.

DETAILED DESCRIPTION

(5) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(6) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “over,” “on,” “top,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(7) FIGS. 1-18B show exemplary sequential processes for manufacturing a semiconductor device structure **100**, in accordance with some embodiments. It is understood that additional operations can be provided before, during, and after processes shown by FIGS. 1-18B, and some of the operations described below can be replaced or eliminated, for additional embodiments of the process. The order of the operations/processes may be interchangeable. FIGS. 19A-22B show alternate sequential processes for manufacturing the semiconductor device structure **100**, in accordance with some embodiments. It is understood that additional operations can be provided before, during, and after processes shown by FIGS. 19A-22B, and some of the operations described below can be replaced or eliminated, for additional embodiments of the process. The order of the operations/processes may be interchangeable.

(8) FIG. 1 is a perspective view of one of the various stages of manufacturing a semiconductor device structure **100**, in accordance with some embodiments. As shown in FIG. 1, the semiconductor device structure **100** includes a substrate **102** having at least a plurality of conductive features **104** formed thereover. The conductive features **104** are formed in a dielectric material **106**. One or more devices, such as transistors, diodes, imaging sensors, resistors, capacitors, inductors, memory cells, a combination thereof, and/or other suitable devices, may be formed between the substrate **102** and the conductive features **104**.

(9) FIGS. 2A-18A are cross-sectional side views of various stages of manufacturing the semiconductor device structure **100** taken along line A-A of FIG. 1, in accordance with some embodiments. FIGS. 2B-18B are cross-sectional side views of various stages of manufacturing the semiconductor device structure **100** taken along line B-B of FIG. 1, in accordance with some embodiments. FIG. 2A is a cross-sectional side view of the semiconductor device structure **100** taken along line A-A of FIG. 1, and FIG. 2B is a cross-sectional side view of the semiconductor device structure **100** taken along line B-B of FIG. 1. The line A-A of FIG. 1 extends along a direction that is substantially perpendicular to the longitudinal direction of a gate stack **206**, and the line B-B of FIG. 1 extends along the longitudinal direction of the gate stack **206**. As shown in FIGS. 2A and 2B, the semiconductor device structure **100** includes the substrate **102**, one or more devices **202** formed on the substrate **102**, the dielectric material **106** formed over the devices **202**, and the conductive features **104** formed in the dielectric material **106**. The substrate **102** may be a semiconductor substrate. In some embodiments, the substrate **102** includes a single crystalline semiconductor layer on at least the surface of the substrate **102**. The substrate **102** may include a single crystalline semiconductor material such as, but not limited to silicon (Si), germanium (Ge), silicon germanium (SiGe), gallium arsenide (GaAs), indium antimonide (InSb), gallium phosphide (GaP), gallium antimonide (GaSb), indium aluminum arsenide (InAlAs), indium gallium arsenide (InGaAs), gallium antimony phosphide (GaSbP), gallium arsenic antimonide (GaAsSb) and indium phosphide (InP). For example, the substrate **102** is made of Si. In some embodiments, the substrate **102** is a silicon-on-insulator (SOI) substrate, which includes an insulating layer (not shown) disposed between two silicon layers. In one aspect, the insulating layer is an oxygen-containing material, such as an oxide.

(10) The substrate **102** may include one or more buffer layers (not shown) on the surface of the substrate **102**. The buffer layers can serve to gradually change the lattice constant from that of the substrate to that of the source/drain regions. The buffer layers may be formed from epitaxially grown single crystalline semiconductor materials such as, but not limited to Si, Ge, germanium tin (GeSn), SiGe, GaAs, InSb, GaP, GaSb, InAlAs, InGaAs, GaSbP, GaAsSb, GaN, GaP, and InP. In one embodiment, the substrate **102** includes SiGe buffer layers epitaxially grown on the silicon substrate **102**. The germanium concentration of the SiGe buffer layers may increase from 30 atomic percent germanium for the bottom-most buffer layer to 70 atomic percent germanium for the top-most buffer layer.

(11) The substrate **102** may include various regions that have been suitably doped with impurities (e.g., p-type or n-type impurities). The dopants are, for example boron for an n-type fin field effect

transistor (FinFET) and phosphorus for a p-type FinFET.

(12) As described above, the devices **202** may be any suitable devices, such as transistors, diodes, imaging sensors, resistors, capacitors, inductors, memory cells, or a combination thereof. In some embodiments, the devices **202** are transistors, such as planar field effect transistors (FETs), FinFETs, nanosheet transistors, or other suitable transistors. The nanosheet transistors may include nanowire transistors, gate-all-around (GAA) transistors, multi-bridge channel (MBC) transistors, or any transistors having the gate electrode surrounding the channels. An example of the device **202** formed between the substrate **102** and the conductive features **104** is a FinFET, which is shown in FIGS. 2A and 2B. The device **202** includes source/drain (S/D) regions **204** and gate stacks **206**. Each gate stack **206** may be disposed between S/D regions **204** serving as source regions and S/D regions **204** serving as drain regions. For example, each gate stack **206** may extend along the Y-axis between a plurality of S/D regions **204** serving as source regions and a plurality of S/D regions **204** serving as drain regions. As shown in FIG. 2A, two gate stacks **206** are formed on the substrate **102**. In some embodiments, more than two gate stacks **206** are formed on the substrate **102**. Channel regions **208** are formed between S/D regions **204** serving as source regions and S/D regions **204** serving as drain regions.

(13) The S/D regions **204** may include a semiconductor material, such as Si or Ge, a III-V compound semiconductor, a II-VI compound semiconductor, or other suitable semiconductor material. Exemplary S/D regions **204** may include, but are not limited to, Ge, SiGe, GaAs, AlGaAs, GaAsP, SiP, InAs, AlAs, InP, GaN, InGaAs, InAlAs, GaSb, AlP, GaP, and the like. The S/D regions **204** may include p-type dopants, such as boron; n-type dopants, such as phosphorus or arsenic; and/or other suitable dopants including combinations thereof. The S/D regions **204** may be formed by an epitaxial growth method using CVD, atomic layer deposition (ALD) or molecular beam epitaxy (MBE). The channel regions **208** may include one or more semiconductor materials, such as Si, Ge, GeSn, SiGe, GaAs, InSb, GaP, GaSb, InAlAs, InGaAs, GaSbP, GaAsSb, GaN, GaP, or InP. In some embodiments, the channel regions **208** include the same semiconductor material as the substrate **102**. In some embodiments, the devices **202** are FinFETs, and the channel regions **208** are located within a plurality of fins disposed below the gate stacks **206**. In some embodiments, the devices **202** are nanosheet transistors, and the channel regions **208** are surrounded by the gate stacks **206**.

(14) Each gate stack **206** includes a gate electrode layer **210** disposed over the channel region **208** (or surrounding the channel region **208** for nanosheet transistors). The gate electrode layer **210** may be a metal-containing material such as tungsten, cobalt, aluminum, ruthenium, copper, multilayers thereof, or the like, and can be deposited by ALD, plasma enhanced chemical vapor deposition (PECVD), MBD, physical vapor deposition (PVD), or any suitable deposition technique. Each gate stack **206** may include an interfacial dielectric layer **212**, a gate dielectric layer **214** disposed on the interfacial dielectric layer **212**, and one or more conformal layers **216** disposed on the gate dielectric layer **214**. The gate electrode layer **210** may be disposed on the one or more conformal layers **216**. The interfacial dielectric layer **212** may include a dielectric material, such as an oxygen-containing material or a nitrogen-containing material, or multilayers thereof, and may be formed by any suitable deposition method, such as CVD, PECVD, or ALD. The gate dielectric layer **214** may include a dielectric material such as an oxygen-containing material or a nitrogen-containing material, a high-k dielectric material having a k value greater than about 7.0, or multilayers thereof. The gate dielectric layer **214** may be formed by any suitable method, such as CVD, PECVD, or ALD. The one or more conformal layers **216** may include one or more barrier layers and/or capping layers, such as a nitrogen-containing material, for example tantalum nitride (TaN), titanium nitride (TiN), or the like. The one or more conformal layers **216** may further include one or more work-function layers, such as aluminum titanium carbide, aluminum titanium oxide, aluminum titanium nitride, or the like. The term “conformal” may be used herein for ease of description upon a layer having substantial same thickness over various regions. The one or more

conformal layers **216** may be deposited by ALD, PECVD, MBD, or any suitable deposition technique.

(15) Gate spacers **218** are formed along sidewalls of the gate stacks **206** (e.g., sidewalls of the gate dielectric layers **214**). The gate spacers **218** may include silicon oxycarbide, silicon nitride, silicon oxynitride, silicon carbon nitride, the like, multi-layers thereof, or a combination thereof, and may be deposited by CVD, ALD, or other suitable deposition technique.

(16) Portions of the gate stacks **206** and the gate spacers **218** may be formed on isolation regions **203**. The isolation regions **203** are formed on the substrate **102**. The isolation regions **203** may include an insulating material such as an oxygen-containing material, a nitrogen-containing material, or a combination thereof. The insulating material may be formed by a high-density plasma chemical vapor deposition (HDP-CVD), a flowable chemical vapor deposition (FCVD), or other suitable deposition process. In one aspect, the isolation regions **203** includes silicon oxide that is formed by a FCVD process.

(17) A contact etch stop layer (CESL) **224** is formed on a portion of the S/D regions **204** and the isolation region **203**, and a first interlayer dielectric (ILD) **226** is formed on the CESL **224**. The CESL **224** can provide a mechanism to stop an etch process when forming openings in the first ILD **226**. The CESL **224** may be conformally deposited on surfaces of the S/D regions **204** and the isolation regions **203**. The CESL **224** may include an oxygen-containing material or a nitrogen-containing material, such as silicon nitride, silicon carbon nitride, silicon oxynitride, carbon nitride, silicon oxide, silicon carbon oxide, or the like, or a combination thereof, and may be deposited by CVD, PECVD, ALD, or any suitable deposition technique. The first ILD **226** may include tetraethylorthosilicate (TEOS) oxide, un-doped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), organosilicate glass (OSG), SiOC, and/or any suitable low-k dielectric materials (e.g., a material having a dielectric constant lower than silicon dioxide), and may be deposited by spin-on, CVD, FCVD, PECVD, PVD, or any suitable deposition technique.

(18) A silicide layer **220** is formed on at least a portion of each S/D region **204**, as shown in FIGS. 2A and 2B. The silicide layer **220** may include a material having one or more of WSi, CoSi, NiSi, TiSi, MoSi and TaSi. In some embodiments, the silicide layer **220** includes a metal or metal alloy silicide, and the metal includes a noble metal, a refractory metal, a rare earth metal, alloys thereof, or combinations thereof. A conductive contact **222** is disposed on each silicide layer **220**. The conductive contact **222** may include a material having one or more of Ru, Mo, Co, Ni, W, Ti, Ta, Cu, Al, TiN or TaN, and the conductive contact **222** may be formed by any suitable method, such as electro-chemical plating (ECP) or PVD. The silicide layer **220** and the conductive contact **222** may be formed by first forming an opening in the first ILD **226** and the CESL **224** to expose at least a portion of the S/D region **204**, then forming the silicide layer **220** on the exposed portion of the S/D region **204**, and then forming the conductive contact **222** on the silicide layer **220**.

(19) An etch stop layer **228** may be formed over the devices **202**, as shown in FIGS. 2A and 2B. The etch stop layer **228** may include the same material as the CESL **224** and may be deposited by the same process as that for the CESL **224**. The dielectric material **106** is formed on the etch stop layer **228**. The dielectric material **106** may be another etch stop layer. The dielectric material **106** may include the same material as the etch stop layer **228** and may be deposited by the same process as that for the etch stop layer **228**. The conductive features **104** are formed in the etch stop layer **228** and the dielectric material **106**, and each conductive feature **104** may be in contact with a corresponding conductive contact **222**.

(20) Next, as shown in FIGS. 3A and 3B, a dielectric material **302** is formed on the dielectric material **106** and the plurality of conductive features **104**. The devices **202** (FIGS. 2A and 2B) and other features formed on the substrate **102** are omitted for clarity. The dielectric material **302** may be a second ILD. The dielectric material **302** may include SiC, LaO, AlO, AlON, ZrO, HfO, SiN, ZnO, ZrN, ZrAlO, TiO, TaO, YO, TaCN, ZrSi, SiOCN, SiOC, SiCN, HfSi, or SiO. In some

embodiments, the dielectric material **302** may have a thickness ranging from about 1 nanometer (nm) to about 40 nm. If the thickness of the dielectric material **302** is greater than about 21 nm, the manufacturing cost is increased without significant advantage. On the other hand, if the thickness of the dielectric material **302** is less than about 1 nm, the dielectric material **302** may be insufficient to isolate any conductive features formed therebelow from the conductive features formed thereon. The dielectric material **302** may be formed by any suitable method, such as CVD or PECVD.

(21) Next, as shown in FIGS. **4A** and **4B**, a mask layer **402** is formed on portions of the dielectric material **302**. The mask layer **402** may be formed by first forming a layer on the dielectric material **302**. The layer may include an oxygen-containing material or a nitrogen-containing material, such as a silicon oxide layer, a silicon nitride layer, a silicon oxynitride layer, or combinations thereof. The layer may be patterned and etched to form the mask layer **402**. The patterning process may include a photolithography process that may include forming a photoresist layer (not shown) over the layer, exposing the resist to a pattern, performing post-exposure bake processes, and developing the resist. In some embodiments, patterning the resist may be performed using an acceptable lithography process, such as an electron beam (e-beam) lithography process, an extreme ultraviolet lithography process, or the like. The pattern of the resist is transferred to the layer using one or more etching processes to form the mask layer **402**. In some embodiments, the etching process may include dry etching (e.g., reactive ion etching (RIE)), wet etching, other etching methods, and/or combinations thereof.

(22) The pattern of the mask layer **402** is transferred to the dielectric material **302** by removing portions of the dielectric material **302** not covered by the mask layer **402**, as shown in FIGS. **5A** and **5B**. The removal of portions of the dielectric material **302** may be performed by any suitable method, such as dry etching, wet etching, or a combination thereof. The mask layer **402** is then removed. The remaining dielectric material **302** has a top surface **503** and sidewalls **504**. Openings **502** are formed as the result of the removal of the portions of the dielectric material **302**. Each opening **502** may be defined by the corresponding sidewall **504**. In some embodiments, openings **502** are trenches, and the sidewall **504** defining each trench includes multiple surfaces, such as 4 surfaces, as shown in FIGS. **5A** and **5B**. In some embodiments, openings **502** are vias, and the sidewall **504** defining each via is a continuous surface. The sidewall **504** may form an acute angle **A** with respect to a top surface **506** of the dielectric material **106** as a result of the etching process. The acute angle **A** may range from about 60 degrees to about 89.5 degrees. In the embodiment where the sidewall **504** includes multiple surfaces, each surface may form the acute angle **A** with respect to the top surface **506** of the dielectric material **106**, the acute angles **A** of the surfaces of the sidewall **504** may be substantially the same or different.

(23) As shown in FIGS. **5A** and **5B**, the openings **502** expose the conductive features **104** and portions of the top surface **506** of the dielectric material **106**. In some embodiments, when the conductive features **104** are not present in the etch stop layer **228** and the dielectric material **106**, portions of the dielectric material **106** and the etch stop layer **228** not covered by the dielectric material **302** are removed to expose the conductive contacts **222** and portions of the first ILD **226** (FIGS. **2A** and **2B**). The removal of the portions of the dielectric material **106** and the etch stop layer **228** may be performed by the same process as the removal of the portion of the dielectric material **302** or by a separate process as the removal of the portion of the dielectric material **302**. As shown in FIG. **5A**, 2 conductive features **104** are exposed along the X-axis. In some embodiments, more than 2 conductive features **104**, such as more than 5 or more than 10 conductive features **104** are exposed. In the embodiment where the conductive features **104** are not present, more than 2 conductive contacts **222** (FIGS. **2A** and **2B**), such as more than 5 or more than 10 conductive contacts **222**, are exposed.

(24) A first barrier layer **602** is formed on the top surfaces **503** of the dielectric material **302**, the sidewalls **504** of the dielectric material **302**, the exposed portions of the top surface **506** of the dielectric material **106**, and the conductive features **104**, as shown in FIGS. **6A** and **6B**. The first

barrier layer **602** may include Co, W, Ru, Al, Mo, Ti, TiN, TiSi, CoSi, NiSi, Cu, TaN, Ni, or TiSiNi. The first barrier layer **602** may be a single layer or a multilayer structure, such as a two-layer structure or a three-layer structure. In some embodiments, the first barrier layer **602** may be conformally deposited and may have a thickness ranging from about 0.5 nm to about 10 nm. The first barrier layer **602** functions as a diffusion barrier layer to prevent a first conductive material **604** from diffusing into the dielectric material **106** and the dielectric material **302**. Thus, if the thickness of the first barrier layer **602** is less than about 0.5 nm, the first barrier layer **602** may not be sufficient to prevent the diffusion of the first conductive material **604** into the dielectric material **106** and the dielectric material **302**. On the other hand, if the thickness of the first barrier layer **602** is greater than about 10 nm, the manufacturing cost is increased without significant advantage. The first barrier layer **602** may be formed by any suitable method, such as ALD, CVD or PECVD.

(25) The first conductive material **604** is formed on the first barrier layer **602**, as shown in FIGS. **6A** and **6B**. The first conductive material **604** may include Co, W, Ru, Al, Mo, Ti, TiN, TiSi, CoSi, NiSi, Cu, TaN, Ni, or TiSiNi. The first conductive material **604** may include the same or different material as the first barrier layer **602**. In some embodiments, the first barrier layer **602** is not present, and the first conductive material **604** is formed on the top surface **503** of the dielectric material **302**, the sidewalls **504** of the dielectric material **302**, the exposed portions of the top surface **506** of the dielectric material **106**, and the conductive features **104**.

(26) Next, as shown in FIGS. **7A** and **7B**, a planarization process is performed to expose the dielectric material **302**. The planarization process may be any suitable process, such as a chemical mechanical polishing (CMP) process. The planarization process removes portions of the first conductive material **604** and portions of the first barrier layer **602** so the first conductive material **604** is substantially coplanar with the dielectric material **302**.

(27) In some embodiments, the first barrier layer **602** and the first conductive material **604** are etched back, as shown in FIGS. **8A** and **8B**. Portions of the first barrier layer **602** disposed on the sidewall **504** are removed to expose a portion of the sidewall **504**. The thickness of the first conductive material **604** is reduced, so the openings **502** are partially filled. The etch back of the first barrier layer **602** and the first conductive material **604** may be performed by any suitable method, such as dry etching, wet etching, or a combination thereof. In some embodiments, a selective dry etching process is utilized to perform the etch back. The selective dry etching process selectively removes portions of the first barrier layer **602** and the first conductive material **604**, while the dielectric material **302** is not removed.

(28) Next, as shown in FIGS. **9A** and **9B**, a second barrier layer **902** is formed on the dielectric material **302**, the exposed portion of the sidewall **504**, the first barrier layer **602**, and the first conductive material **604**. The second barrier layer **902** may include Co, W, Ru, Al, Mo, Ti, TiN, TiSi, CoSi, NiSi, Cu, TaN, Ni, or TiSiNi. The second barrier layer **902** may be a single layer or a multilayer structure, such as a two-layer structure or a three-layer structure. In some embodiments, the second barrier layer **902** may be conformally deposited and may have a thickness ranging from about 0.5 nm to about 10 nm. The second barrier layer **902** functions as a diffusion barrier layer to prevent a second conductive material **904** from diffusing into the dielectric material **302**. Thus, if the thickness of the second barrier layer **902** is less than about 0.5 nm, the second barrier layer **902** may not be sufficient to prevent the diffusion of the conductive material **904** into the dielectric material **302**. On the other hand, if the thickness of the second barrier layer **902** is greater than about 10 nm, the manufacturing cost is increased without significant advantage. The second barrier layer **902** may be formed by any suitable method, such as ALD, CVD or PECVD.

(29) The second conductive material **904** is formed on the second barrier layer **902**, as shown in FIGS. **9A** and **9B**. The second conductive material **904** may include Co, W, Ru, Al, Mo, Ti, TiN, TiSi, CoSi, NiSi, Cu, TaN, Ni, or TiSiNi. The second conductive material **904** may include the same or different material as the second barrier layer **902**. In some embodiments, the second barrier layer **902** is not present, and the second conductive material **904** is formed on the dielectric

material **302**, the exposed portion of the sidewall **504**, the first barrier layer **602**, and the first conductive material **604**.

(30) In some embodiments, the first conductive material **604** is a metal having a low electrical resistivity, such as copper, and the first conductive material **604** fills the openings **502** without the second conductive material **904** (the second conductive material **904** and the second barrier layer **902** are not present). However, as the dimensions of features get smaller, materials such as copper may not have good step coverage in the openings **502**. Thus, in some embodiments, a conductive material having good step coverage in the openings **502** may be utilized as the first conductive material **604**, and a conductive material having low electrical resistivity may be utilized as the second conductive material **904**. The bottom of the opening **502** has a smaller dimension than the top of the opening **502**, thus, the first conductive material **604** having good step coverage is formed at the bottom of the opening **502** and the second conductive material **904** having low electrical resistivity is formed at the top of the opening **502**. For example, the first conductive material **604** is TiN and the second conductive material **904** is Cu. In some embodiments, the thickness of the first conductive material **604** ranges from about 0.5 nm to about 40 nm, and the thickness of the second conductive material **904** ranges from about 0.5 nm to about 38 nm. In some embodiments, the second conductive material **904** does not exist, and the thickness of the first conductive material **604** ranges from about 0.5 nm to about 40 nm. The thicknesses of the first conductive material **604** and the second conductive material **904** may be defined by the thickness of the dielectric material **302**.

(31) Next, as shown in FIGS. **10A** and **10B**, a planarization process is performed to expose the dielectric material **302**. The planarization process may be any suitable process, such as a CMP process. The planarization process removes portions of the second conductive material **904** and portions of the second barrier layer **902** so the second conductive material **904** is substantially coplanar with the dielectric material **302**. The first barrier layer **602**, the first conductive material **604**, the second barrier layer **902**, and the second conductive material **904** may be collectively referred to as a conductive structure **1002**. The conductive structure **1002** may be a conductive contact, a conductive line, or a conductive via. In some embodiments, the conductive structure **1002** includes the first conductive material **604** and optionally the first barrier layer **602**, while the second barrier layer **902** and the second conductive material **904** are not present. The conductive structure **1002** includes a sidewall **1004** in contact with the sidewall **504** of the dielectric material **302**. Because the sidewall **1004** is in contact with the sidewall **504** of the dielectric material **302**, the sidewall **1004** also forms the acute angle A with respect to the top surface **506** of the dielectric material **106**. The sidewall **1004** of the conductive structure **1002** may include one or more of the first barrier layer **602**, first conductive material **604**, second barrier layer **902**, and second conductive material **904**. For example, in some embodiments, the sidewall **1004** includes the first barrier layer **602** and the second barrier layer **902**, as shown in FIGS. **10A** and **10B**. The sidewall **1004** of the conductive structure **1002** may include multiple surfaces or a continuous surface, based on the shape of the conductive structure **1002**. In some embodiments, the conductive structure **1002** is a conductive line, and the sidewall **1004** of the conductive line includes multiple surfaces, such as 4 surfaces, as shown in FIGS. **10A** and **10B**. In some embodiments, the conductive structure **1002** is a conductive via, and the sidewall **1004** of the conductive via is a continuous surface. The conductive structure **1002** may have a first width extending along the Y-axis ranging from about 3 nm to about 15 nm at the top and a second width extending along the Y-axis ranging from about 3 nm to about 15 nm at the bottom. In some embodiments, the first width is greater than the second width.

(32) In some embodiments, the dielectric material **302** is etched back to form air gaps **1106** between neighboring conductive structures **1002**, as shown in FIGS. **11A** and **11B**. The thickness of the dielectric material **302** is reduced to a range from about 0.5 nm to about 20 nm, and a first portion **1102** of the sidewall **1004** of the conductive structure **1002** is exposed. The thickness of the

dielectric material **302** is reduced, so a spacer layer **1202** (FIGS. **12A** and **12B**) may be conformally formed on the dielectric material **302** with the reduced thickness and on the sidewall **1004** of the conductive structure **1002**. Thus, if the thickness of the dielectric material **302** is greater than about 20 nm, the portion of the spacer layer **1202** (FIGS. **12A** and **12B**) disposed on the dielectric material **302** and the portion of the spacer layer **1202** (FIGS. **12A** and **12B**) disposed on the sidewall **1004** of the conductive structure **1002** may not provide enough room for the air gap **1106**, leading to small openings **1204** (FIGS. **12A** and **12B**) of the air gaps **1106**. As a result, the dimensions of the air gap **1106** may be too small to achieve any device performance gain. In some embodiments, as shown in FIGS. **11A** and **11B**, the dielectric material **302** is etched back. In some embodiments, as shown in FIGS. **20A** and **20B**, the dielectric material **302** is removed.

(33) The first portion **1102** of the sidewall **1004** of the conductive structure **1002** may include a portion of the second barrier layer **902** and a portion of the first barrier layer **602**, as shown in FIGS. **11A** and **11B**. In some embodiments, the first portion **1102** of the sidewall **1004** of the conductive structure **1002** includes one or more of the first barrier layer **602**, first conductive material **604**, second barrier layer **902**, and second conductive material **904**. The etch back of the dielectric material **302** may be performed by any suitable method, such as dry etching, wet etching, or a combination thereof. In some embodiments, a selective dry etching process is utilized to perform the etch back. The selective dry etching process selectively removes a portion of the dielectric material **302**, while the conductive structure **1002** is not removed.

(34) As shown in FIGS. **11A** and **11B**, the dielectric material **302** surrounds a second portion **1104** of the sidewall **1004** of each conductive structure **1002**. The second portion **1104** of the sidewall **1004** of the conductive structure **1002** may include a portion of the first barrier layer **602**, as shown in FIGS. **11A** and **11B**. In some embodiments, the second portion **1104** of the sidewall **1004** of the conductive structure **1002** includes one or more of the first barrier layer **602** and first conductive material **604**. In some embodiments, the first portion **1102** is a top portion of the sidewall **1004** of the conductive structure **1002**, which is disposed on the second portion **1104**, which is a bottom portion of the sidewall **1004** of the conductive structure **1002**, as shown in FIGS. **11A** and **11B**.

(35) Next, as shown in FIGS. **12A** and **12B**, the spacer layer **1202** is formed on the dielectric material **302**, the sidewall **1004** (the first barrier layer **602** and the second barrier layer **902**), and the conductive structure **1002**. The opening **1204** of the air gap **1106** may be defined by the spacer layer **1202**. The spacer layer **1202** may include SiC, LaO, AlO, AlON, ZrO, HfO, SiN, Si, ZnO, ZrN, ZrAlO, TiO, TaO, YO, TaCN, ZrSi, SiOCN, SiOC, SiCN, HfSi, or SiO. The spacer layer **1202** may include a different material as the dielectric material **302**, and the spacer layer **1202** and the dielectric material **302** may have different etch selectivity. In some embodiments, the spacer layer **1202** may be conformally deposited in the air gaps **1106** and may have a thickness ranging from about 0.5 nm to about 6 nm. The thickness range of the spacer layer **1202** defines the opening **1204** of the air gap **1106**. Thus, if the thickness of the spacer layer **1202** is less than about 0.5 nm, the opening **1204** may be too large, any material, such as a sealing material **1502** (FIGS. **15A** and **15B**), formed over the air gaps **1106** may fill the air gaps **1106**. On the other hand, if the thickness of the spacer layer **1202** is greater than about 6 nm, the opening **1204** may be too small, and the air gap **1106** may be too small to provide improved isolation between neighboring conductive structures **1002**. The spacer layer **1202** may be formed by any suitable method, such as ALD, CVD or PECVD.

(36) Next, as shown in FIGS. **13A** and **13B**, portions of the spacer layer **1202** are removed. In some embodiments, the portions of the spacer layer **1202** disposed on the dielectric material **302** and the conductive structure **1002** are removed, leaving the portion of the spacer layer **1202** adjacent and in contact with the sidewall **1004** of the conductive structure **1002**. The spacer layer **1202** adjacent and in contact with the sidewall **1004** may have a height along the Z-axis ranging from about 0.5 nm to about 35 nm. The height of the spacer layer **1202** may be defined by the thicknesses of the dielectric material **302** before the etch back and after the etch back. The dielectric material **302** and

the top of the conductive structure **1002** are exposed.

(37) The removal of the portions of the spacer layer **1202** may be performed by any suitable method, such as an etching process. In one example, the etching process is an anisotropic selective dry etch process. The anisotropic selective dry etch removes the portions of the spacer layer **1202** disposed on horizontal surfaces but does not remove the portions disposed adjacent and in contact with the sidewalls **1004** of the conductive structure **1002**. The anisotropic selective dry etch process selectively removes the portions of the spacer layer **1202**, while the dielectric material **302** and the conductive structure **1002** are not removed.

(38) Next, as shown in FIGS. **14A** and **14B**, the dielectric material **302** surrounding the second portion **1104** of the sidewall **1004** of the conductive structure **1002** is removed to expose the second portion **1104** of the sidewall **1004** of the conductive structure **1002**. The removal of the dielectric material **302** may be performed by any suitable method, such as an etching process. In one example, the etching process is an isotropic selective dry etch process that removes the dielectric material **302**, while the spacer layer **1202** and the conductive structure **1002** are not removed. The spacer layer **1202** is in contact with the first portion **1102** of the sidewall **1004**, while the second portion **1104** of the sidewall **1004** is exposed to the air gap **1106**. The distance between the bottom of the spacer layer **1202** and the dielectric material **106** may range from about 0.5 nm to about 6 nm, and the distance is defined by the thickness of the dielectric material **302** previously surrounding the second portion **1104** of the sidewall **1004**.

(39) The sealing material **1502** is formed on the spacer layers **1202** and the conductive structures **1002**, as shown in FIGS. **15A** and **15B**. The sealing material **1502** may also seal the air gaps **1106** by partially fill the air gaps **1106**. The sealing material **1502** does not completely fill the air gaps **1106** due to the small opening **1204** (FIGS. **12A** and **12B**) of the air gap **1106**. The sealing material **1502** may include SiC, LaO, AlO, AlON, ZrO, HfO, SiN, Si, ZnO, ZrN, ZrAlO, TiO, TaO, YO, TaCN, ZrSi, SiOCN, SiOC, SiCN, HfSi, or SiO. The sealing material **1502** may include the same or different material as the spacer layer **1202**. The sealing material **1502** may be formed by any suitable method, such as CVD.

(40) Next, as shown in FIGS. **16A** and **16B**, a planarization process is performed to expose the conductive structure **1002** and the spacer layer **1202**. The planarization process may be any suitable process, such as a CMP process. The planarization process removes portions of the sealing material **1502** so the remaining sealing material **1502** disposed over the air gaps **1106** is substantially coplanar with the conductive structure **1002**. As described above, the sealing material **1502** partially fills the air gap **1106**. As a result, the air gap **1106** has a height **H1** ranging from about 0.5 nm to about 30 nm, and the sealing material **1502** has a height **H2** ranging from about 0.5 nm to about 20 nm. The height **H1** may be defined by the thickness of the dielectric layer **302** before being etched back and by the height **H2**. The height **H2** may be defined by the size of the opening **1204**, which is defined by the thickness of the spacer layer **1202**. The air gap **1106** may be defined by the dielectric material **106**, the second portion **1104** of the sidewall **1004** of neighboring conductive structure **1002**, the neighboring spacer layers **1202**, and the sealing material **1502**. In some embodiments, the second portion **1104** of the sidewall **1004** of a first conductive structure **1002** and a portion of the spacer layer **1202** disposed adjacent and in contact with the first portion **1102** of the sidewall of the first conductive structure **1002** are exposed to a first air gap **1106**. The second portion **1104** of the sidewall **1004** of a second conductive structure **1002** adjacent the first conductive structure **1002** and a portion of the spacer layer **1202** disposed adjacent and in contact with the first portion **1102** of the sidewall of the second conductive structure **1002** are exposed to the first air gap **1106**.

(41) In some embodiments, a width **W** of the air gap **1106** along the Y-axis varies based on the height **H1** of the air gap **1106**. In one aspect, the width **W** decreases in the direction of the height **H1** moving away from the dielectric material **106**. For example, the width **W** at the top of the air gap is **W1** (FIG. **17A**), which may range from about 3 nm to about 16 nm. The range of the width

W1 may be defined by the height H2 of the sealing material. The width W at the bottom of the air gap **1106** is W2 (FIG. **17B**), which may range from about 3 nm to about 30 nm. The width W2 at the bottom of the air gap **1106** may be defined by the arrangement of the conductive structures **1002**, which in turn may be defined by the arrangements of the conductive features **104**. The width W may be generally defined by the arrangements of the conductive structures **1002**. Thus, the lower limit of 3 nm may be defined by the pitch of conductive structures **1002**, not feasible to go any lower. On the other hand, if the distance between conductive structures **1002** are greater than 30 nm, capacitive coupling between the conductive structures **1002** may be low, rendering the air gap **1106** formed therebetween a result of increasing manufacturing cost without significant advantage.

(42) FIG. **17A** is a top view of the semiconductor device structure **100** at the manufacturing stage taken along line C-C as shown in FIG. **16A**, in accordance with some embodiments. As shown in FIG. **17A**, the air gap **1106** surrounds the spacer layer **1202**, which surrounds the first portion **1102** of the sidewall **1004** of the conductive structure **1002**. The air gap **1106** is a continuous air gap that surrounds multiple surfaces, such as 4 surfaces of the sidewall **1004** of the conductive structure **1002**. The width W1 of the air gap **1106** between neighboring spacer layers **1202** may range from about 3 nm to about 16 nm. The air gaps **1106** may replace the dielectric material **302** (FIGS. **10A** and **10B**). In other words, because the dielectric material **302** may be the second ILD, the second ILD may be replaced by the air gaps **1106**. The air gap **1106**, which has a lower k value compared to the materials of the spacer layer **1202** and the dielectric material **302**, is formed to isolate conductive structures **1002**, leading to reduced capacitive coupling between neighboring conductive structures **1002**.

(43) FIG. **17B** is a top view of the semiconductor device structure **100** at the manufacturing stage taken along line D-D as shown in FIG. **16A**, in accordance with some embodiments. As shown in FIG. **17B**, the air gap **1106** surrounds the second portion **1104** of the sidewall **1004** of the conductive structure **1002**. The width W2 of the air gap **1106** between neighboring second portions **1104** of the sidewalls **1004** may range from about 3 nm to about 30 nm. In some embodiments, the width W2 is greater than the width W1.

(44) A dielectric material **1802** is formed on the sealing materials **1502**, the spacer layers **1202**, and the conductive structures **1002**, as shown in FIGS. **18A** and **18B**. The dielectric material **1802** may be a third ILD. The dielectric material **1802** may include SiC, LaO, AlO, AlON, ZrO, HfO, SiN, ZnO, ZrN, ZrAlO, TiO, TaO, YO, TaCN, ZrSi, SiOCN, SiOC, SiCN, HfSi, or SiO. The dielectric material **1802** may include the same or different material as the dielectric material **302**. In some embodiments, the sealing material **1502** includes the same material as the dielectric material **1802**, and the sealing material **1502** may function as the third ILD. In such embodiments, the planarization process described in FIGS. **16A** and **16B** and the deposition of the dielectric material **1802** described in FIGS. **18A** and **18B** are skipped. Conductive structures (not shown) may be formed in the dielectric material **1802** to connect to the conductive structures **1002**. In some embodiments, the dielectric material **1802** may be replaced by air gaps based on the processes described above.

(45) FIGS. **19A** and **19B** are cross-sectional side views of the semiconductor device structure **100** at the manufacturing stage right after FIGS. **10A** and **10B**, in accordance with some embodiments. As shown in FIGS. **19A** and **19B**, instead of etching back the dielectric material **302** as shown in FIGS. **11A** and **11B**, the dielectric material **302** is removed. Thus, both first portion **1102** and second portion **1104** of the sidewall **1004** of the conductive structure **1002** are exposed. The spacer layer **1202** is formed on the dielectric material **106**, the first portion **1102**, the second portion **1104**, and the conductive structure **1002**, as shown in FIGS. **20A** and **20B**. Next, similar to the processes described in FIGS. **13A**, **13B**, **15A**, and **15B**, portions of the spacer layer **1202** are removed and the sealing material **1502** is formed on the conductive structures **1002** and partially fill the air gaps **1106**, as shown in FIGS. **21A** and **21B**. The spacer layer **1202** disposed adjacent and in contact with

the sidewall **1004** extends to and in contact with the dielectric material **106**, as shown in FIGS. **21A** and **21B**. The spacer layer **1202** may surround the sidewall **1004** of the conductive structure **1002**, and the air gap **1106** surrounds the spacer layer **1202**. In some embodiments, the air gap **1106** may be defined by the dielectric material **106**, the neighboring spacer layers **1202**, and the sealing material **1502**. As shown in FIG. **22B**, the width **W2** at the bottom of the air gap **1106** may range from about 2 nm to about 18 nm.

(46) Next, as shown in FIGS. **22A** and **22B**, portions of the sealing material **1502** are removed, and the dielectric material **1802** is formed on the sealing materials **1502** and the conductive structures **1002**, similar to the processes described in FIGS. **16A**, **16B**, **18A**, and **18B**.

(47) The present disclosure provides a semiconductor device structure **100** including a device **202**, a conductive structure **1002** disposed above the device **202**, a spacer layer **1202** disposed on at least a portion of the sidewall **1004** of the conductive structure **1002**, and an air gap **1106** surrounding the spacer layer **1202**. Some embodiments may achieve advantages. For example, the spacer layer **1202** defines the opening **1204** of the air gap **1106**, so the air gap **1106** can provide improved isolation between neighboring conductive structures **1002** while preventing materials from filling the air gaps **1106**. The air gap **1106** has a lower k value compared to the spacer layer **1202**, which reduces capacitive coupling between neighboring conductive structures **1002**.

(48) An embodiment is a semiconductor device structure. The semiconductor device structure includes a device, a first conductive structure disposed over the device, and the first conductive structure includes a first sidewall having a first portion and a second portion. The semiconductor device structure further includes a first spacer layer disposed on the first portion of the first sidewall of the first conductive structure, a second conductive structure disposed adjacent the first conductive structure, and the second conductive structure includes a second sidewall having a third portion and a fourth portion. The semiconductor device structure further includes a second spacer layer disposed on the third portion of the second sidewall of the second conductive structure, and an air gap is formed between the first conductive structure and the second conductive structure. The second portion of the first sidewall of the first conductive structure, the first spacer layer, the fourth portion of the second sidewall of the second conductive structure, and the second spacer layer are exposed to the air gap.

(49) Another embodiment is a semiconductor device structure. The semiconductor device structure includes a device, a first dielectric material disposed over the device, a first conductive structure disposed over the first dielectric material, and the first conductive structure includes a first sidewall. The semiconductor structure device further includes a first spacer layer disposed on the first sidewall of the first conductive structure, a second conductive structure disposed adjacent the first conductive structure, and the second conductive structure includes a second sidewall. The semiconductor device structure further includes a second spacer layer disposed on the second sidewall of the second conductive structure, and a sealing material disposed between the first spacer layer and the second spacer layer. An air gap is defined by the first dielectric material, the first spacer layer, the second spacer layer, and the sealing material.

(50) A further embodiment is a method. The method includes forming a device over a substrate, forming a dielectric material over the device, forming a first opening and a second opening in the dielectric material, and forming a first conductive structure in the first opening and a second conductive structure in the second opening. The first conductive structure includes a first sidewall having a first portion and a second portion, and the second conductive structure includes a second sidewall having a third portion and a fourth portion. The method further includes removing at least a portion of the dielectric material between the first conductive structure and the second conductive structure, and the first portion of the first sidewall of the first conductive structure and the third portion of the second sidewall of the second conductive structure are exposed. The method further includes forming a first spacer layer on the first portion of the first sidewall of the first conductive structure and a second spacer layer on the third portion of the second sidewall of the second

conductive structure, and forming a sealing material between the first and second conductive structures. An air gap is formed between the first conductive structure and the second conductive structure, and the sealing material, the first spacer layer, and the second spacer layer are exposed to the air gap.

(51) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device structure, comprising: a first dielectric material disposed over a substrate; a first conductive structure disposed over the first dielectric material, wherein the first conductive structure comprises a first sidewall, and the first sidewall comprises a conductive material; a second conductive structure disposed adjacent the first conductive structure, wherein the second conductive structure comprises a second sidewall; and a sealing material disposed between the first conductive structure and a second conductive structure, wherein an air gap is formed between the first and second conductive structures, a portion of the first sidewall and a portion of the second sidewall are exposed to the air gap, and the air gap extends from the portion of the first sidewall to the portion of the second sidewall.
2. The semiconductor device structure of claim 1, wherein the first conductive structure further comprises a first conductive feature.
3. The semiconductor device structure of claim 2, wherein the first conductive structure further comprises a first barrier layer, and wherein the first conductive feature is disposed on the first barrier layer.
4. The semiconductor device structure of claim 3, wherein the first conductive structure further comprises a second barrier layer disposed on the first conductive feature and a second conductive feature disposed on the second barrier layer.
5. The semiconductor device structure of claim 4, wherein the portion of the first sidewall comprises a portion of the first barrier layer.
6. The semiconductor device structure of claim 1, wherein the air gap has a varying width.
7. The semiconductor device structure of claim 6, wherein the width of the air gap decreases in a direction away from the substrate.
8. A semiconductor device structure, comprising: a first conductive structure disposed over a substrate, wherein the first conductive structure comprises a first sidewall; a first spacer layer disposed on the first sidewall of the first conductive structure, wherein a top surface of the first spacer layer and a top surface of the first conductive structure are substantially co-planar; a second conductive structure disposed adjacent the first conductive structure, wherein the second conductive structure comprises a second sidewall facing the first spacer layer; and a second spacer layer disposed on the second sidewall of the second conductive structure, wherein a top surface of the second spacer layer and a top surface of the second conductive structure are substantially co-planar, and an air gap is formed between the first and second spacer layers, wherein the air gap extends from the first spacer layer to the second spacer layer.
9. The semiconductor device structure of claim 8, further comprising a first dielectric material, wherein the first and second conductive structures are disposed over the dielectric material.
10. The semiconductor device structure of claim 9, wherein the first spacer layer and the second

spacer layer are in contact with the first dielectric material.

11. The semiconductor device structure of claim 10, further comprising a sealing material disposed between the first and second spacer layers, wherein the air gap is defined by the first and second spacer layers, the first dielectric material, and the sealing material.

12. The semiconductor device structure of claim 11, further comprising a second dielectric material disposed on the sealing material, the first conductive structure, and the second conductive structure.

13. The semiconductor device structure of claim 11, wherein a top surface of the sealing material and the top surface of the first spacer layer are substantially co-planar.

14. The semiconductor device structure of claim 8, wherein the air gap has a varying width.

15. The semiconductor device structure of claim 14, wherein the width of the air gap decreases in a direction away from the substrate.

16. A structure, comprising: a first dielectric material disposed over a substrate; a first conductive structure disposed over the first dielectric material, wherein the first conductive structure comprises a first sidewall including a first portion and a second portion, and the first portion is disposed over the second portion; a first spacer layer disposed on the first portion of the first sidewall of the first conductive structure; and a sealing material in contact with the first spacer layer, wherein an air gap is located below the sealing material, wherein the first dielectric material, the first spacer layer, the second portion of the first side wall, and the sealing material are exposed to the air gap.

17. The structure of claim 16, further comprising a second conductive structure disposed adjacent the first conductive structure, wherein the second conductive structure comprises a second sidewall including a third portion and a fourth portion.

18. The structure of claim 17, further comprising a second spacer layer disposed on the third portion of the second sidewall of the second conductive structure, wherein the fourth portion of the second sidewall of the second conductive structure and the second spacer layer are exposed to the air gap.

19. The structure of claim 16, wherein the air gap surrounds the first conductive structure.

20. The structure of claim 16, wherein the air gap has a varying width.
